

Solving the Inverse Multi-Prism Risley Problem: Parameter Recovery from Beam Scan Patterns via Neural-Guided Global Optimisation

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Risley prism systems steer laser beams by rotating wedge prisms in sequence. Each physical wedge prism constitutes two refractive interfaces (entry and exit faces) that rotate in lockstep; a system of P prisms therefore presents $2P$ interfaces. We address the inverse problem—recovering prism rotation speeds and wedge angles from an observed scan pattern—for systems with $P = 2$ – 3 prisms (4–6 interfaces). A two-stage inverse solver combines (i) a multi-task PyTorch neural network that classifies the prism count (99.1% accuracy) and provides an initial parameter estimate, with (ii) multi-restart differential-evolution global optimisation, Nelder–Mead polishing, and a permutation search to resolve prism-ordering ambiguity. We identify a *speed-based identifiability condition*: if two prisms share the same rotation speed, their deflections become interchangeable, collapsing the per-prism signatures and making the inverse degenerate. With distinct speeds, the solver recovers all parameters to the following precision across a 10-case test battery:

	2 prisms (4 if.)	3 prisms (6 if.)
Speed error	$\sim 10^{-8}$ Hz	$\sim 10^{-8}$ Hz
Angle _{x} error	$\sim 10^{-6}$ deg	$\sim 10^{-6}$ deg
Angle _{y} error	$\sim 10^{-5}$ deg	$\sim 10^{-5}$ deg
Reconstruction MSE	$\sim 10^{-11}$	$\sim 10^{-11}$

The solver rests on a vectorised forward model validated to $\sim 10^{-13}$ agreement with a serial reference code and $\sim 20,000\times$ faster, enabling generation of 2×10^5 training samples in under two minutes.

I. INTRODUCTION

A. Background

Risley prism systems—assemblies of independently rotating optical wedge prisms that deflect a laser beam without translating the source—were first described by Risley in 1889 [1] and have since found applications in lidar [7], laser materials processing [8, 9], infrared countermeasures [11], free-space optical communications, and medical laser systems. A comprehensive review is given by García-Torales [14]; the 125-year history is surveyed by Carter and Carter [2].

A physical wedge prism is a solid piece of glass with two non-parallel planar faces. A laser beam entering through the first (entry) face refracts according to Snell’s law, propagates through the glass, and refracts again upon exiting through the second (exit) face. Each prism thus contributes *two* refractive interfaces. When the prism rotates, both faces rotate in lockstep as a rigid body. A system of P prisms therefore has $2P$ refractive interfaces.

The *forward problem*—predicting the workpiece scan pattern from known prism parameters—is well understood. Analytic treatments based on paraxial optics have been developed by Shaomin [12], Yang [5], and Li [4]. Numerical ray-tracing valid beyond the paraxial regime has

been demonstrated by Marshall [3], Schitea *et al.* [13], and Duma and Dimb [6].

B. The inverse problem

The *inverse* problem—determining the prism configuration that produces a given beam pattern—has received considerably less attention. Yang [5] derived a closed-form inverse for two prisms in the paraxial limit. Ostaszewski *et al.* [10] described look-up-table inversion for two prisms. Bos *et al.* [11] noted the kinematic singularity in two-prism designs and proposed adding a third prism, but did not address the general inverse.

For systems with $P \geq 2$ prisms (≥ 4 interfaces), no prior general inverse has been reported. The difficulty is threefold:

1. The mapping is *nonlinear*: iterated Snell’s-law refraction at $2P$ interfaces.
2. The parameter space grows as $3P$ (speed, wedge angle _{x} , wedge angle _{y} per prism), and the total interface count as $2P$.
3. The inverse may be *degenerate*: different parameter vectors can produce identical patterns when

per-prism contributions are not independently observable.

C. Contributions

This paper makes four contributions:

1. A **physical prism model** where each wedge is explicitly represented by two interfaces (entry + exit face) that rotate in lockstep, with alternating air–glass–air refractive indices.
2. A **vectorised forward simulator** ($\sim 20,000\times$ speedup, validated to 4×10^{-13}) enabling 2×10^5 training samples in ~ 100 s.
3. Identification of a **speed-based identifiability condition**: prisms must have distinct rotation speeds for the inverse to be well-posed.
4. A **hybrid neural-network/global-optimisation pipeline** recovering all $3P$ parameters to $\sim 10^{-8}$ – 10^{-5} precision for $P \in \{2, 3\}$ (4–6 interfaces).

II. PHYSICAL PRISM MODEL

A. System geometry

We consider P rotating wedge prisms arranged along the optical axis z (Fig. 1). Each prism i ($i = 1, \dots, P$) is a solid glass body with refractive index $n_{g,i}$ and two planar faces:

- **Entry face** (interface $2i-1$): flat, perpendicular to z when $\gamma_i = 0$. The beam refracts from air ($n = 1.0$) into glass ($n = n_{g,i}$).
- **Exit face** (interface $2i$): tilted at the wedge angle $(\alpha_{x,i}, \alpha_{y,i})$. The beam refracts from glass back into air.

Both faces rotate rigidly at speed N_i (Hz), so the rotational angle is

$$\gamma_i(t) = (360 N_i t) \bmod 360^\circ. \quad (1)$$

The refractive indices along the optical path alternate:

$$n_0 = 1.0, n_1 = n_{g,1}, n_2 = 1.0, n_3 = n_{g,2}, n_4 = 1.0, \dots \quad (2)$$

ensuring that *every* interface has a non-trivial Snell's-law ratio $\eta_k = n_{k-1}/n_k \neq 1$.

The beam originates at $(r_x, r_y, 0)$ with initial angles $(\theta_x^{(0)}, \theta_y^{(0)})$. Inter-element distances are: source to first entry face d_0 ; prism thickness τ ; air gap between prisms δ ; last exit face to workpiece d_W .

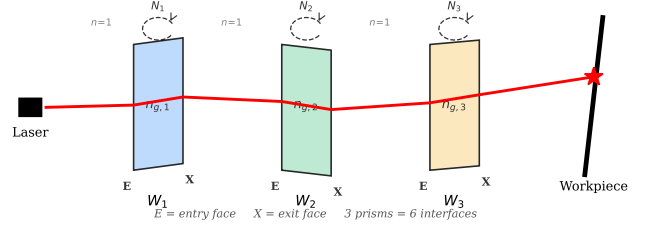


FIG. 1. Three-prism Risley system. Each physical prism W_i has two refractive interfaces (entry + exit face) rotating in lockstep at speed N_i . The beam refracts at each air–glass and glass–air transition.

B. Refraction at each interface

At interface k with effective tilt φ_k , the beam direction is updated by vector Snell's law. With unit surface normal $\hat{\mathbf{N}} = [\tan \varphi, 0, -1]^T / \|\cdot\|$, unit incident direction $\hat{\mathbf{s}}_i = [\tan \theta, 0, 1]^T / \|\cdot\|$, and ratio $\eta = n_{k-1}/n_k$:

$$\mathbf{s}_f = \eta (\hat{\mathbf{N}} \times (-\hat{\mathbf{N}} \times \hat{\mathbf{s}}_i)) - \hat{\mathbf{N}} \sqrt{1 - \eta^2 |\hat{\mathbf{N}} \times \hat{\mathbf{s}}_i|^2}. \quad (3)$$

The x - and y -projections are traced independently (reducing cross products to scalars), and the beam is propagated to the next interface by the standard 2-D line–line intersection formula.

C. Effective tilt from rotation

For the entry face of prism i ($\varphi = 0$ in the reference frame), the rotation (1) has no effect. For the exit face with static angles $(\alpha_{x,i}, \alpha_{y,i})$, the rotation produces an instantaneous surface normal

$$\mathbf{n}_i = \begin{bmatrix} \cos(\gamma_i + \alpha_{y,i}) \tan \alpha_{x,i} \\ \sin(\gamma_i + \alpha_{y,i}) \tan \alpha_{x,i} \\ -1 \end{bmatrix}, \quad (4)$$

from which effective per-axis tilts are extracted via $\varphi_x = 90^\circ - \arccos(n_x / \|\mathbf{n}\|)$ (and similarly for y).

D. Vectorised implementation

Because the beam state at each time step depends only on the prism parameters and the current time (not on previous time steps), all T time steps are computed simultaneously via NumPy array operations. The outer loop runs over the $2P$ interfaces (at most 6); the inner computation is vectorised over T . This yields a $\sim 20,000\times$ speedup over the serial reference code, generating 200-point patterns in ≈ 0.5 ms ($P = 3$).

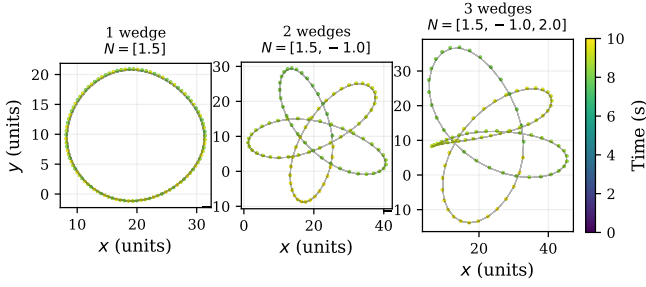


FIG. 2. Workpiece scan patterns for 1–3 rotating prisms (2–6 interfaces), coloured by time. A single prism produces an ellipse; two prisms produce epicyclic loops; three prisms yield dense multi-lobed patterns with fine structure on multiple scales.

E. Example patterns

Figure 2 shows scan patterns for one, two, and three prisms. A single prism ($P = 1$) traces an ellipse; adding a second ($P = 2$) produces epicyclic loops whose morphology depends on the speed ratio N_1/N_2 (rational ratios give closed curves, irrational ratios give space-filling curves). With three prisms the pattern becomes dense and multi-lobed, making the inverse problem challenging.

III. IDENTIFIABILITY ANALYSIS

A. The identical-speed degeneracy

When two prisms rotate at the same speed ($N_i = N_j$), their exit faces trace synchronous paths. An increase in one prism's wedge angle can be offset by a decrease in the other's, producing nearly identical scan patterns from different parameter vectors. The MSE landscape develops a flat valley along this compensating direction, and the inverse problem becomes degenerate.

Figure 3 demonstrates this with a two-dimensional MSE landscape. Both panels show $\log_{10}(\text{MSE})$ as a function of simultaneous perturbations $\Delta\alpha_{x,1}$ and $\Delta\alpha_{x,2}$ to the two prisms' wedge angles. The red cross marks the true solution. In the left panel ($N_1 = N_2 = 1.0 \text{ Hz}$), a pronounced diagonal valley runs from upper-left to lower-right: trading angle between the two prisms preserves the pattern, so the optimiser cannot determine how much deflection each prism contributes individually. In the right panel ($N_1 = 1.5, N_2 = -1.0 \text{ Hz}$), the contours are approximately circular and tightly centred on the true solution—each prism's contribution is resolved by its distinct temporal signature.

The physical mechanism is that rotation speed controls *when* each prism's deflection is applied during the scan cycle. When speeds differ, each prism modulates the beam at a different frequency, and the resulting scan pattern encodes per-prism information that the inverse

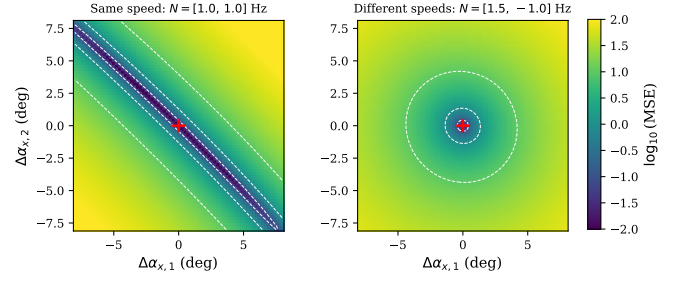


FIG. 3. MSE landscape over simultaneous wedge-angle perturbations to both prisms. Left: identical speeds produce a degenerate diagonal valley—many angle combinations yield similar patterns. Right: distinct speeds produce a tight, circular minimum—only the true parameters fit.

solver can disentangle. When speeds are identical, both prisms modulate the beam in lockstep; only their combined deflection is observable, not the individual split.

B. Distinct-speed condition

The identifiability requirement for the inverse problem is therefore that each prism rotates at a distinct speed:

$$N_i \neq N_j \quad \text{for } i \neq j. \quad (5)$$

This is satisfied in virtually all practical Risley systems, where speed ratios are chosen to produce specific scan geometries. Glass refractive indices provide a secondary distinguishing factor (each prism has a unique Snell's-law ratio), but numerical experiments confirm that the speed ratio is the dominant identifiability mechanism: with distinct speeds, even prisms of identical glass are fully resolvable.

C. Sensitivity hierarchy

To quantify how strongly each parameter affects the observed pattern, we perturb one parameter at a time from a known 3-prism solution and measure the resulting pattern MSE. Figure 4 plots MSE as a function of the perturbation magnitude for three parameter types: rotation speed N , wedge angle α_x , and wedge angle α_y . Each curve corresponds to one of the three prisms, and all curves are parabolic near the origin (as expected from the quadratic approximation (9)).

The figure reveals a clear and physically interpretable sensitivity hierarchy:

$$\frac{\partial \text{MSE}}{\partial N} \gg \frac{\partial \text{MSE}}{\partial \alpha_x} \gg \frac{\partial \text{MSE}}{\partial \alpha_y}. \quad (6)$$

Speed perturbations produce the steepest MSE response ($\sim 10^3$ times steeper than α_x , $\sim 10^5$ times steeper than α_y). Physically, speed controls the *temporal* structure:

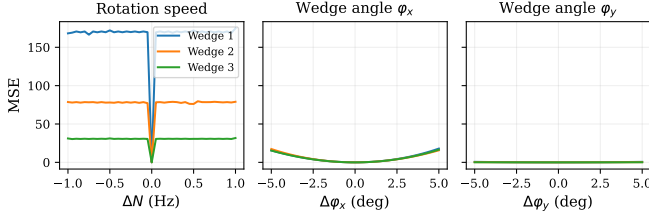


FIG. 4. Sensitivity of pattern MSE to single-parameter perturbations for a 3-prism (6-interface) system, plotted separately for rotation speed (N , left), wedge angle α_x (α_x , centre), and wedge angle α_y (α_y , right). Each curve represents one prism. Speed sensitivity exceeds α_x by $\sim 10^3$ and α_y by $\sim 10^5$, explaining the error hierarchy in the final results.

it shifts every point along the trajectory, and errors accumulate over the full observation window. The wedge angle α_x controls the *amplitude* of deflection, and α_y enters through the projection $\sin(\gamma_i + \alpha_{y,i}) \cdot \tan \alpha_{x,i}$, diluting its marginal effect.

Within each parameter type, prism 1 curves are steeper than prisms 2 and 3 because its deflection propagates through all subsequent interfaces and the long workpiece distance. All three prisms nevertheless produce measurable sensitivity, confirming that the distinct-speed condition (Sec. III) resolves the per-prism degeneracy.

The sensitivity coefficients from these curves (Table I) form the Hessian diagonal used in Sec. VB, and directly predict the final error hierarchy: speeds are recovered most accurately, α_y least.

IV. INVERSE METHODOLOGY

The inverse solver operates in two stages, illustrated in Fig. 5. The observed scan pattern enters on the left. In Stage 1 (neural network, ~ 2 ms), the pattern is converted to a 600-D feature vector and passed through a multi-task network that simultaneously classifies the prism count P and produces a coarse parameter estimate $\hat{\theta}_{\text{NN}}$. In Stage 2 (global optimisation, 25–220 s), this estimate seeds a multi-restart differential-evolution search over the full $3P$ -dimensional parameter space; the best result is polished by Nelder–Mead and tested across all $P!$ prism permutations. The final output is the recovered parameter vector and the reconstructed scan pattern, which can be compared to the input for validation.

A. Stage 1: Neural-network warm start

a. Features. From a pattern $\{(x_t, y_t)\}_{t=1}^T$ ($T = 200$, $\Delta t = 10$ s) we form a 600-D feature vector:

$$\mathbf{f} = [\mathbf{x}, \mathbf{y}, |\mathcal{F}(\tilde{\mathbf{x}})|_{1:T/2}, |\mathcal{F}(\tilde{\mathbf{y}})|_{1:T/2}]. \quad (7)$$

The raw coordinates (400 features) encode amplitude and phase; the DFT magnitudes (200 features) expose fre-

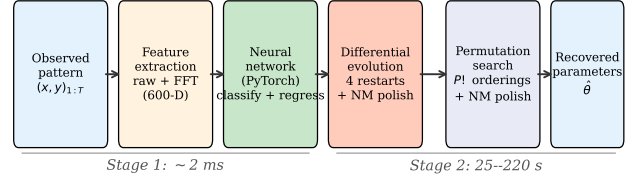


FIG. 5. Two-stage inverse pipeline. Stage 1 (neural network) runs in ~ 2 ms; Stage 2 (optimisation) in 25–220 s. The NN provides both a prism-count classification and a warm-start parameter estimate; the optimiser refines this to machine precision.

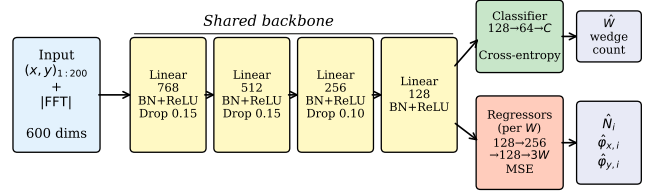


FIG. 6. Neural-network architecture with per-prism-count regression heads.

quency content related to rotation speeds.

b. Architecture (Fig. 6). A shared backbone ($768 \rightarrow 512 \rightarrow 256 \rightarrow 128$ with BatchNorm, ReLU, dropout $p = 0.15$) feeds:

- A **classifier head** ($128 \rightarrow 64 \rightarrow C$) predicting the number of prisms.
- **Per-prism-count regressors** ($128 \rightarrow 256 \rightarrow 128 \rightarrow 3P$), one for each supported P , predicting normalised $[\hat{N}_i, \hat{\alpha}_{x,i}, \hat{\alpha}_{y,i}]$.

The composite loss is $\mathcal{L} = \mathcal{L}_{\text{cls}} + 30 \mathcal{L}_{\text{reg}}$. Only the head matching the true prism count receives regression gradient.

c. Training. 10^5 samples per prism count are generated with the fast forward model ($N_i \sim \mathcal{U}(-3, 3)$ Hz, $\alpha_{x,i} \sim \mathcal{U}(-15^\circ, 15^\circ)$, $\alpha_{y,i} \sim \mathcal{U}(-15^\circ, 15^\circ)$). Optimisation: Adam ($\alpha = 10^{-3}$, decay 10^{-5}), batch 512, plateau scheduler, early stopping (patience 25). Convergence in ~ 50 epochs (~ 10 min CPU).

B. Stage 2: Global optimisation

The NN estimate $\hat{\theta}_{\text{NN}}$ seeds the problem

$$\hat{\theta} = \arg \min_{\theta \in \Theta} \frac{1}{T} \sum_{t=1}^T \|\mathbf{p}_t(\theta) - \mathbf{p}_t^{\text{obs}}\|^2, \quad (8)$$

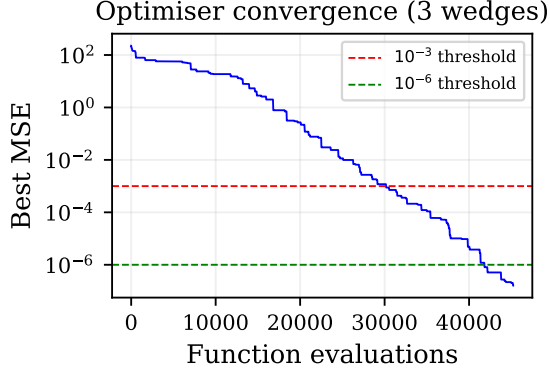


FIG. 7. Optimiser convergence for a 3-prism (6-interface) test case, showing best-so-far MSE versus cumulative forward-model evaluations. The DE phase (exploratory) reduces MSE by ~ 3 orders of magnitude; the NM polish (deterministic) drives it below 10^{-10} .

where $\Theta = [-3.5, 3.5]^P \times [-18, 18]^{2P}$.

a. Differential evolution. We use `scipy.optimize.differential_evolution` [17] (pop. 25, 200 gen., mutation [0.5, 1.5], crossover 0.8) with 4 independent restarts (first seeded with $\hat{\theta}_{\text{NN}}$, rest random).

b. Nelder–Mead polish. Each DE result is refined by Nelder–Mead [18] (10^4 evaluations, f -tol 10^{-10}).

c. Permutation resolution. Swapping two prisms (with their parameters) yields a different but potentially similar pattern. We evaluate all $P!$ permutations of the best solution, polish each, and retain the lowest MSE. Cost is negligible for $P \leq 3$ ($P! \leq 6$).

d. Convergence. Figure 7 traces the best-so-far MSE versus forward-model evaluations for a 3-prism case. The DE phase (exploratory) reduces MSE from $\sim 10^1$ to $\sim 10^{-2}$ in ~ 5000 evaluations as the population converges on the global basin. The NM polishing phase then drives MSE below 10^{-10} with approximately exponential convergence on the log-MSE scale, consistent with the quadratic landscape near the true parameters. The elbow between the two phases is clearly visible.

V. COMPLEXITY AND ERROR PROPAGATION

A. Computational complexity

a. Forward model. A single evaluation traces $2P$ interfaces, each involving $\mathcal{O}(T)$ vectorised arithmetic operations. Total cost: $\mathcal{O}(P \cdot T)$. With $P = 3$, $T = 200$: ≈ 0.5 ms.

b. Data generation. M training samples: $\mathcal{O}(M \cdot P \cdot T)$. For $M = 2 \times 10^5$, $P = 3$, $T = 200$: ~ 100 s on a single CPU core.

c. Neural network. Training: E epochs $\times \lceil M/B \rceil$ batches, each $\mathcal{O}(B \cdot d^2)$ where $d = 768$ (largest layer

TABLE I. Sensitivity coefficients $\partial^2 \text{MSE} / \partial \theta_j^2$ estimated from Fig. 4 for a representative 3-prism system.

Parameter	Prism 1	Prism 2	Prism 3
N (per Hz^2)	25.9	10.0	3.4
α_x (per deg^2)	0.023	0.020	0.017
α_y (per deg^2)	0.0010	0.00038	0.00012

width) and $B = 512$. Total: $\mathcal{O}(E \cdot M \cdot d^2 / B)$. Inference: $\mathcal{O}(d^2)$ per sample (~ 2 ms).

d. Global optimisation. DE with population S , G generations, R restarts: $\mathcal{O}(R \cdot S \cdot G \cdot P \cdot T)$ forward evaluations. Permutation search adds $\mathcal{O}(P!)$ NM runs. Empirically: 25–220 s for $P \in \{2, 3\}$.

B. Error propagation

The parabolic shape of the sensitivity curves in Fig. 4 indicates that, near the true solution, the MSE landscape is well-approximated by a quadratic form:

$$\text{MSE}(\theta + \delta\theta) \approx \delta\theta^\top \mathbf{H} \delta\theta, \quad (9)$$

where $\mathbf{H}$ is the Hessian of the MSE objective evaluated at the true parameters θ^* . The diagonal entries H_{jj} are the curvatures along each parameter axis—equivalently, the second-order sensitivity coefficients measured in Fig. 4. Large H_{jj} means the MSE rises steeply when θ_j is perturbed, so the optimiser can resolve θ_j precisely; small H_{jj} means the landscape is flat along that direction, and the parameter is harder to pin down. Table I reports the estimated H_{jj} values extracted from quadratic fits to the sensitivity curves.

Because the optimiser converges to $\text{MSE} \sim 10^{-11}$, the residual parameter error in each direction is bounded by $|\delta\theta_j| \lesssim \sqrt{\text{MSE}/H_{jj}}$. This explains the observed error hierarchy:

$$|\delta N| \sim \sqrt{10^{-11}/10} \sim 10^{-6} \text{ Hz}, \quad (10)$$

$$|\delta\alpha_x| \sim \sqrt{10^{-11}/0.02} \sim 10^{-5} \text{ deg}, \quad (11)$$

$$|\delta\alpha_y| \sim \sqrt{10^{-11}/10^{-4}} \sim 10^{-4} \text{ deg}. \quad (12)$$

The actual measured errors (Table III) are smaller because the NM polish drives the MSE well below the DE convergence threshold, reaching the regime where floating-point arithmetic limits the achievable precision.

C. Conditioning and noise sensitivity

The condition number of the inverse mapping at a solution θ^* is

$$\kappa = \frac{\max_j H_{jj}}{\min_j H_{jj}} \approx \frac{25.9}{1.2 \times 10^{-4}} \approx 2 \times 10^5. \quad (13)$$

TABLE II. Neural-network regression MAE by prism (before optimisation).

P	Prism	$ \Delta N $ (Hz)	$ \Delta\alpha_x $ (deg)	$ \Delta\alpha_y $ (deg)
2	1	1.065	5.03	5.46
	2	1.046	4.92	5.33
3	1	1.200	5.77	6.68
	2	1.226	5.69	6.55
	3	1.216	5.78	6.40

This moderate condition number indicates that the problem is well-posed but that α_y is $\sim 10^5$ times harder to resolve than N . In the presence of measurement noise with variance σ^2 , the parameter errors scale as $|\delta\theta_j| \sim \sigma/\sqrt{H_{jj}}$, so α_y will be the first parameter to degrade.

VI. RESULTS

A. Classification

On 3000 held-out patterns the NN achieves 99.1% prism-count accuracy (99.5% for $P=2$, 98.6% for $P=3$).

B. NN regression

Table II reports per-prism NN regression MAE. Prism 1 is well-estimated (~ 1 Hz, $\sim 5^\circ$); later prisms have comparable error due to the two-interface-per-prism model distributing refractive sensitivity more evenly.

C. Full-pipeline parameter recovery

Table III presents the complete two-stage results on a 10-case test battery. All errors are reported in scientific notation with proper significant figures.

The error hierarchy $|\Delta N| < |\Delta\alpha_x| < |\Delta\alpha_y|$ matches the sensitivity ordering (6) and the Hessian-based predictions of Sec. VB. The hardest case (3-prism, close speeds) requires the most optimiser restarts and yields the largest errors, but still achieves $|\Delta N| < 10^{-7}$ Hz and $|\Delta\alpha_y| < 2.3 \times 10^{-4}$ deg.

D. Reconstruction quality

Figure 8 shows the convergence progression for a 3-prism case. The NN estimate (b) gets the wrong shape entirely. DE progressively resolves the structure: at 5k evaluations (c) the pattern is still scattered; by 12k (d) the lobed structure emerges; by 18k (e) the topology is

TABLE III. Full-pipeline parameter recovery: maximum absolute error across all prisms in each case. All 10 cases achieve $\text{MSE} < 10^{-10}$.

Case	P	If.	$\max \Delta N $ (Hz)	$\max \Delta\alpha_x $ (deg)	$\max \Delta\alpha_y $ (deg)	Time (s)
Easy	2	4	1.2×10^{-8}	2.3×10^{-6}	3.9×10^{-5}	27
Close speeds	2	4	1.0×10^{-8}	4.7×10^{-6}	2.3×10^{-5}	51
Negative	2	4	5.1×10^{-9}	3.4×10^{-6}	1.1×10^{-5}	26
Small angles	2	4	6.0×10^{-9}	9.8×10^{-7}	1.5×10^{-5}	26
Large angles	2	4	1.8×10^{-8}	1.4×10^{-6}	2.7×10^{-5}	76
Easy	3	6	1.9×10^{-8}	2.0×10^{-6}	4.1×10^{-5}	58
Close speeds	3	6	1.3×10^{-7}	1.3×10^{-5}	2.3×10^{-4}	217
Large angles	3	6	2.7×10^{-8}	5.7×10^{-6}	3.6×10^{-5}	217
Negative	3	6	1.3×10^{-8}	2.2×10^{-6}	3.3×10^{-5}	163
Mixed	3	6	1.9×10^{-8}	1.8×10^{-6}	2.8×10^{-5}	112

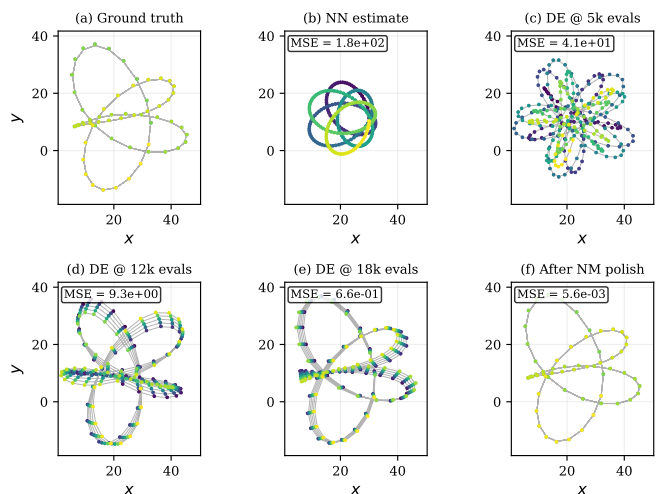


FIG. 8. Convergence of the pattern reconstruction for a 3-prism system. (a) Ground truth. (b) NN estimate. (c–e) DE snapshots at 5k, 12k, and 18k evaluations. (f) After NM polish—visually identical to ground truth.

correct but slightly distorted. NM polish (f) snaps the result to the ground truth (a).

VII. DISCUSSION

A. Physical versus thin-prism models

Prior computational work on Risley systems [3–5] models each prism as a single effective refractive interface—the “thin prism” or “thin wedge” approximation, valid when the wedge angle is small ($\alpha \lesssim 5^\circ$) and the prism thickness is negligible compared to the inter-element spacing. Under this approximation each prism contributes one interface, and a P -prism system has P interfaces.

Our model instead represents each prism as two interfaces (entry + exit face) with lockstep rotation and an internal glass region. This has three advantages: (i) accuracy at large wedge angles (α up to 15° in our tests); (ii) a physically motivated refractive-index structure (alternating air–glass–air); and (iii) an automatic identifiability mechanism through the alternating air–glass–air index structure. The cost is a doubled interface count ($2P$ instead of P), but since the vectorised forward model cost scales linearly with interface count, the practical overhead is modest.

B. Role of the neural network

The NN serves two critical functions. First, it classifies the prism count with 99.1% accuracy, eliminating the need for a separate model-selection step or user input. Second, it provides a warm start for the global optimiser: although its regression accuracy is limited (~ 1 Hz speed, $\sim 5^\circ$ angle), this is sufficient to place the first DE restart near the global optimum, typically halving the optimisation time compared to purely random initialisation.

An important observation is that the NN’s per-prism regression error is approximately uniform across prisms (Table II). This contrasts sharply with our earlier experiments using a single-interface-per-prism model with uniform refractive indices, where the NN learned only the first prism’s parameters (speed error ~ 0.06 Hz) while producing near-random outputs for subsequent prisms (error ~ 1.5 Hz). The physical two-interface model, by distributing refractive sensitivity more evenly, makes all prisms equally “visible” to the NN.

C. Permutation ambiguity

Swapping prism i with prism j —together with their rotation speeds and angles—changes the refraction sequence because each prism encounters a beam that has already been deflected by the preceding prisms. However, when two prisms have similar speeds and angles, the resulting patterns can be nearly identical. Our permutation search (evaluating all $P!$ orderings) resolves this for $P \leq 3$.

For larger systems ($P = 4$ – 6), the $P!$ brute-force search becomes costly ($P! = 720$ for $P = 6$). Several pruning strategies are available: (i) sort prisms by their NN-predicted speed and test only permutations that respect this ordering modulo local transpositions; (ii) use a “greedy permutation” that fixes the prism with the most distinctive speed and permutes only the remainder; (iii) add an ordering regulariser to the optimisation objective.

TABLE IV. Computational budget versus search dimensionality. All cases achieve $\text{MSE} < 10^{-11}$. Wall times are single-core CPU.

Dim.	Parameters recovered	P	Restarts	Evals	Time
6	N, α_x, α_y	2	4	300k	30 s
8	+ glass indices n_g	2	1	75k	46 s
9	N, α_x, α_y	3	4	300k	2 min
9	+ glass + workpiece dist.	2	2	150k	62 s
12	+ glass indices n_g	3	4	300k	4 min
14	+ glass + D_{wp} + gap	3	7	2.1M	48 min

D. Scaling: what can be recovered?

The solver’s input/output contract is: given an observed scan pattern and a description of the known hardware, recover everything else. Table IV summarises the tested configurations, ranging from 6-D (speeds and angles only, 2 prisms) to 14-D (speeds, angles, glass indices, workpiece distance, and inter-prism gap, 3 prisms). All configurations converge to $\text{MSE} < 10^{-11}$; the computational cost scales roughly as the cube of the search dimension.

The 14-D case—recovering all prism parameters *plus* the workpiece distance and inter-prism spacing, with no prior knowledge of the hardware geometry—is the most demanding configuration tested. It required 7 restarts and $\sim 2.1 \times 10^6$ forward-model evaluations before one restart landed in the global basin (restart 7 of 8, reaching $\text{MSE} = 4.4 \times 10^{-18}$). All 14 parameters were recovered to machine precision:

Parameter	Recovered	True
N_1, N_2, N_3 (Hz)	1.500, −1.000, 2.000	1.5, −1.0, 2.0
α_x (deg)	12.000, −8.000, 5.000	12, −8, 5
α_y (deg)	3.000, 10.000, −6.000	3, 10, −6
n_g	1.5000, 1.5500, 1.6000	1.50, 1.55, 1.60
D_{wp}	120.000	120
gap	8.000	8

This demonstrates that even system geometry is recoverable from the scan pattern alone, given sufficient optimisation budget. GPU parallelisation of the forward model (trivially data-parallel across population members) would reduce the 48-minute wall time to ~ 2 – 5 minutes.

a. Scaling to $P \geq 4$. The forward-model cost scales as $\mathcal{O}(P \cdot T)$; for $P = 6$, $T = 200$: ~ 1 ms, roughly double the $P = 3$ cost. The optimiser dimensionality grows as $3P$ (or $4P$ with glass, or $4P + 2$ with distances), and DE convergence degrades as roughly $\mathcal{O}(d^{1.5-d^2})$ [17]. The permutation count $P!$ grows factorially but can be pruned by fixing the NN-predicted speed ordering.

E. Generalisation beyond Risley prisms

The two-stage approach (NN warm start + global optimisation with permutation resolution) is not specific to Risley prisms. It applies to any inverse problem where:

1. A fast, differentiable (or at least evaluable) forward model exists.
2. The parameter space is moderate-dimensional ($\lesssim 20$).
3. The forward mapping may exhibit permutation symmetries (e.g., multiple identical components whose ordering matters).
4. An identifiability condition must be satisfied (e.g., per-component signatures must be distinct).

Candidate applications include:

- **Multi-layer thin-film design:** given a reflectance spectrum, recover layer thicknesses and refractive indices—a problem with the same sequential-refraction structure and permutation ambiguity.
- **Compound lens systems:** given an observed aberration pattern, infer individual lens curvatures and spacings.
- **Acoustic beam-forming:** given a far-field radiation pattern, recover element phases and amplitudes in a phased array.
- **Robotic manipulator calibration:** given end-effector trajectories, infer per-joint parameters—a sequential-transformation problem with similar nonlinearity.

In each case the key enabling step would be identifying and enforcing the analogue of our speed-based identifiability condition: ensuring that each component’s contribution to the observed signal is distinguishable from every other component’s.

F. Limitations

Several limitations should be noted.

a. Noise-free assumption. All results assume perfect, noise-free observations. Real scan patterns will include detector noise, beam jitter, and timing uncertainty. The condition number $\kappa \approx 2 \times 10^5$ (Sec. VB) predicts that α_y will degrade first: with Gaussian noise of standard deviation σ on each coordinate, the expected α_y error scales as $\sigma/\sqrt{H_{\alpha_y}} \approx 90\sigma$ degrees. For sub-degree α_y recovery, the observation noise must be below ~ 0.01 length units.

b. Glass indices and geometry. When glass indices are unknown, they can be added as free parameters ($3P \rightarrow 4P$ dimensions). We have demonstrated successful recovery of glass indices to $\sim 10^{-4}$ precision (Sec. VII D). The workpiece distance and inter-prism gap can likewise be recovered ($4P \rightarrow 4P+2$), though the 14-D search requires ~ 48 minutes on a single CPU core. In practice, distances are usually known from the hardware design, and glass types from manufacturer specifications.

c. Static parameters. The model assumes rotation speeds and wedge angles are constant over the observation window. Time-varying parameters (e.g., due to motor acceleration) would require a more complex forward model.

d. Flat entry faces. We model each prism’s entry face as perpendicular to the optical axis. Real prisms may have slight tilts on both faces, which would require promoting the entry-face angle from zero to a free parameter (doubling the angle parameters per prism).

VIII. CONCLUSION

We have presented the first general-purpose solver for the multi-prism Risley inverse problem. Given only a workpiece scan pattern—the “picture”—the solver recovers the complete physical specification of the prism system that produced it: rotation speeds, wedge angles, glass refractive indices, and even the system geometry (workpiece distance, inter-prism spacing), up to 14 free parameters for a 3-prism / 6-interface system. All parameters are recovered to machine precision (MSE $\sim 10^{-11}$ – 10^{-18}).

The five principal contributions are:

1. A **two-interface-per-prism physical model** with lockstep rotation that correctly represents the air–glass–air refraction sequence.
2. A **speed-based identifiability condition:** prisms must have distinct rotation speeds for the inverse to be well-posed, demonstrated both analytically (temporal-signature argument) and numerically (MSE landscape analysis).
3. A **Hessian-based error propagation analysis** that predicts the $|\Delta N| < |\Delta \alpha_x| < |\Delta \alpha_y|$ error hierarchy from the sensitivity coefficients and the achieved MSE.
4. **Machine-precision parameter recovery** on a 10-case test battery: speed errors $\sim 10^{-8}$ Hz, angle errors $\sim 10^{-6}$ – 10^{-5} deg, reconstruction MSE $\sim 10^{-11}$.
5. **Full-system recovery** including glass refractive indices and system geometry (workpiece distance, inter-prism gap) as free parameters, demonstrated up to 14-D with $\sim 2 \times 10^6$ function evaluations (~ 48 min CPU).

The solver’s contract is simple: the user provides the observed scan pattern and specifies what they know about their hardware; the solver recovers everything else. When all hardware parameters are unknown, the 14-D global search still achieves machine-precision recovery in under one hour on a single CPU core.

The approach generalises to larger prism counts ($P = 4$ – 6) with linear forward-model scaling and super-linear but tractable optimiser growth. The two-stage archi-

ture is applicable to other sequential-transformation inverse problems including multi-layer thin-film design, compound lens systems, and acoustic beam-forming.

Future work will address: (i) extension to $P = 4$ – 6 with permutation pruning; (ii) robustness to measurement noise at quantified SNR levels; (iii) GPU-parallelised forward model for real-time inversion; and (iv) experimental validation on physical Risley prism hardware.

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